

- RECAP:
- 1) Stationarity: $\{X_t : t \in \mathcal{T}\}$ a random process w.r.t $(\Omega, \mathcal{F}, \mathbb{P})$
 - a) Strict Sense: $\mathbb{P}_{t_1, \dots, t_n}(x_1, \dots, x_n) = \mathbb{P}_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$, $\forall n \in \mathbb{N}$ and $\forall \{t_1, \dots, t_n\} \subset \mathcal{T}$
Translation-invariance
 - b) N^{th} order: $\mathbb{P}_{t_1, \dots, t_n}(x_1, \dots, x_n) = \mathbb{P}_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$ $\forall n \leq N, n \in \mathbb{N}$ and $\forall \{t_1, \dots, t_n\} \subset \mathcal{T}$
 - c) Wide Sense: $\mu_X(t)$ is a constant
 $R_X(s, t)$ is a function of $s-t$ only, $\mathbb{E}[X_t^2] < \infty \forall t \in \mathcal{T}$.

- PLAN:
- 1) IID Process: Bernoulli Process.
 - 2) Arrival and Renewal processes.
 - 3) Poisson Process.

Reference: Bruce Hajek, "Random Processes for Engineers".
 R. G. Gallager, "Stochastic Processes Theory for Applications"

IID Processes

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\{X_t : t \in \mathcal{T}\}$ be a random process defined w.r.t $\mathcal{F}$

Defn: $\{X_t : t \in \mathcal{T}\}$ is said to be an **IID Process** with the **common CDF F** if for any $n \in \mathbb{N}$ and $\{t_1, t_2, \dots, t_n\} \subset \mathcal{T}$, $\mathbb{P}_{t_1, t_2, \dots, t_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n F(x_i)$, $x_1, \dots, x_n \in \mathbb{R}$

i.e. For any $n \in \mathbb{N}$ and $\{t_1, t_2, \dots, t_n\} \subset \mathcal{T}$, $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ are i.i.d random variables.

- If $\{X_t : t \in \mathcal{T}\}$ is an IID process:
- i) the FDDs of $\{X_t : t \in \mathcal{T}\}$ are consistent
 - ii) $\{X_t : t \in \mathcal{T}\}$ is strictly stationary

Every IID Process is Stationary!

Bernoulli Process :

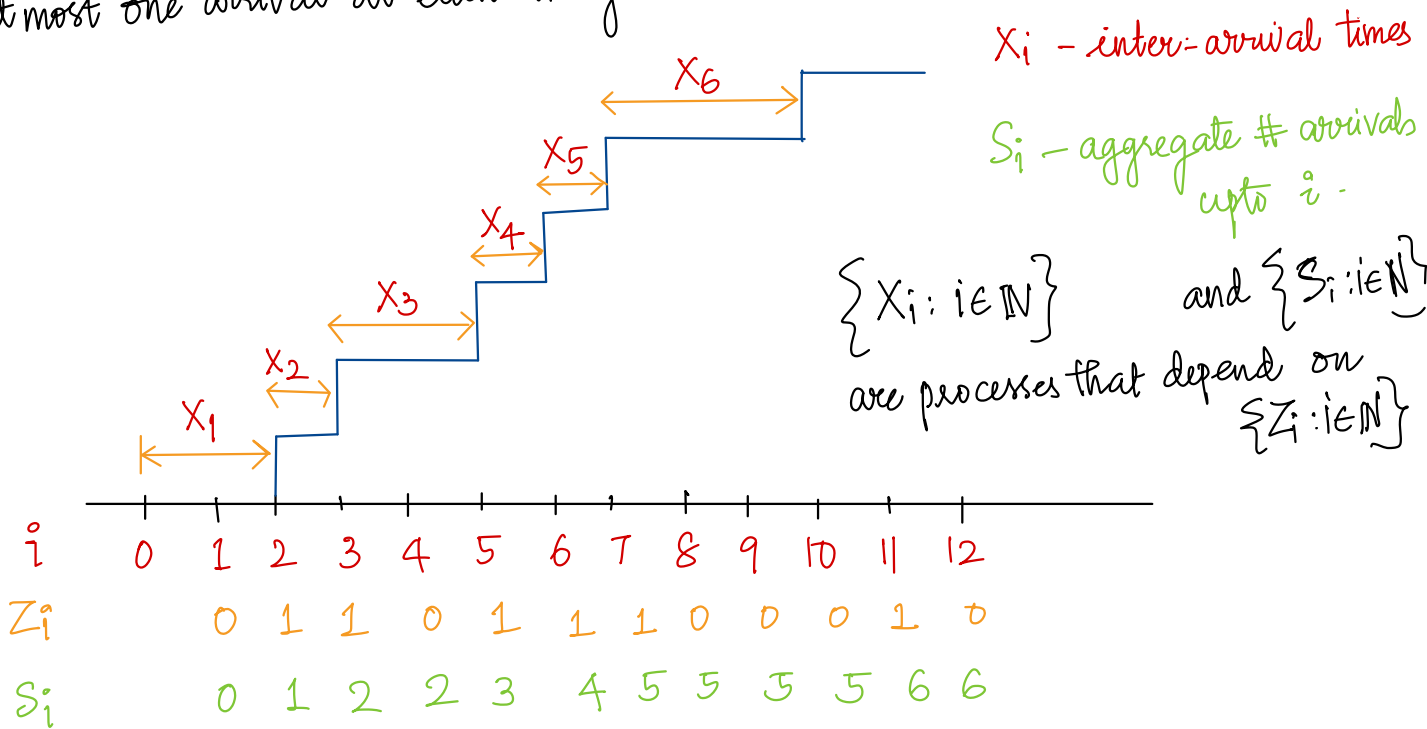
The Bernoulli process is a sequence $Z_1, Z_2, \dots$ of iid binary RVs.

Qn: When is an infinite sequence of RVs independent?

Let $p = \Pr(Z_i = 1)$ and $q = 1-p = \Pr(Z_i = 0)$.

Bernoulli Process : evolving in discrete time with the event $\{Z_i = 1\}$ representing a customer arriving at instant i and $\{Z_i = 0\} \Rightarrow$ no arrival.

i.e., at most one arrival at each integer instant.



$$X_1 = 1 \text{ iff } Z_1 = 1 \Rightarrow P_{X_1}(1) = p.$$

$$X_1 = 2 \text{ iff } Z_1 = 0 \text{ and } Z_2 = 1 \Rightarrow P_{X_1}(2) = (1-p)p.$$

$$X_1 \sim \text{Geo}(p)$$

$$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Geo}(p)$$

$$X_2 \sim \text{Geo}(p)$$

$\therefore$ Sequence of inter-arrival times is also an IID process with geometric distribution.

A sample path of $\{X_i: i \in \mathbb{N}\}$ determines a sample path of $\{Z_i: i \in \mathbb{N}\}$.

$\Rightarrow$ Bernoulli process can be characterised by a sequence of geometric RVs.

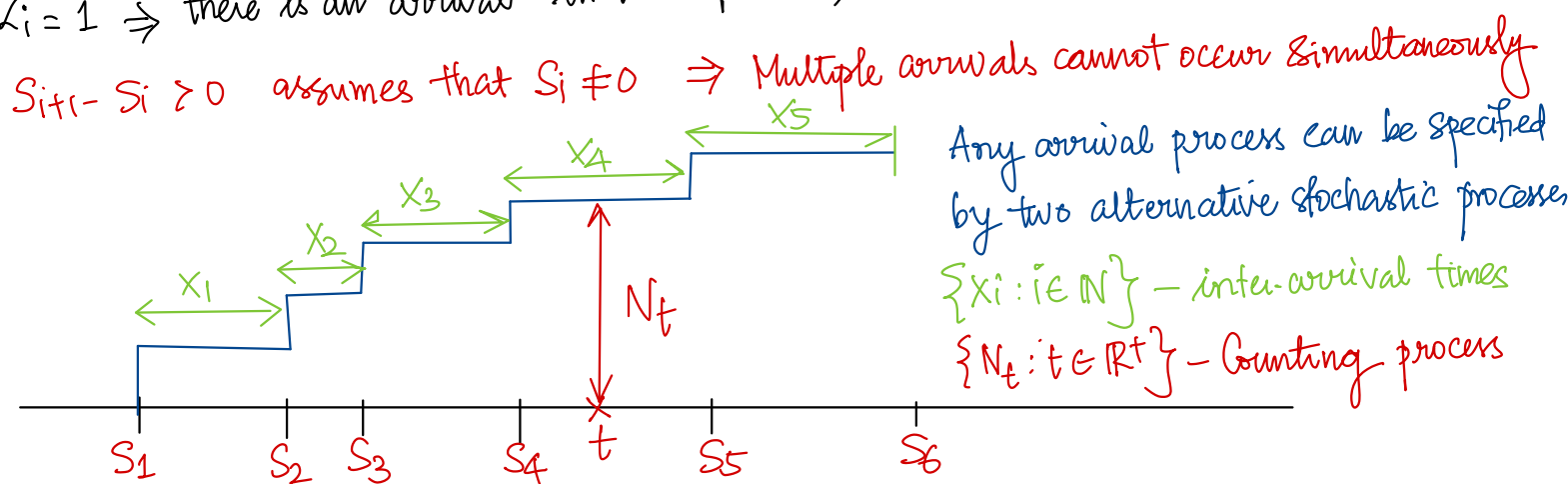
$\{S_n: n \in \mathbb{N}\}$ — where $S_n = \sum_{i=1}^n Z_i$ — number of arrivals upto and including the time instant n .

$$S_n \sim \text{Bin}(n, p)$$

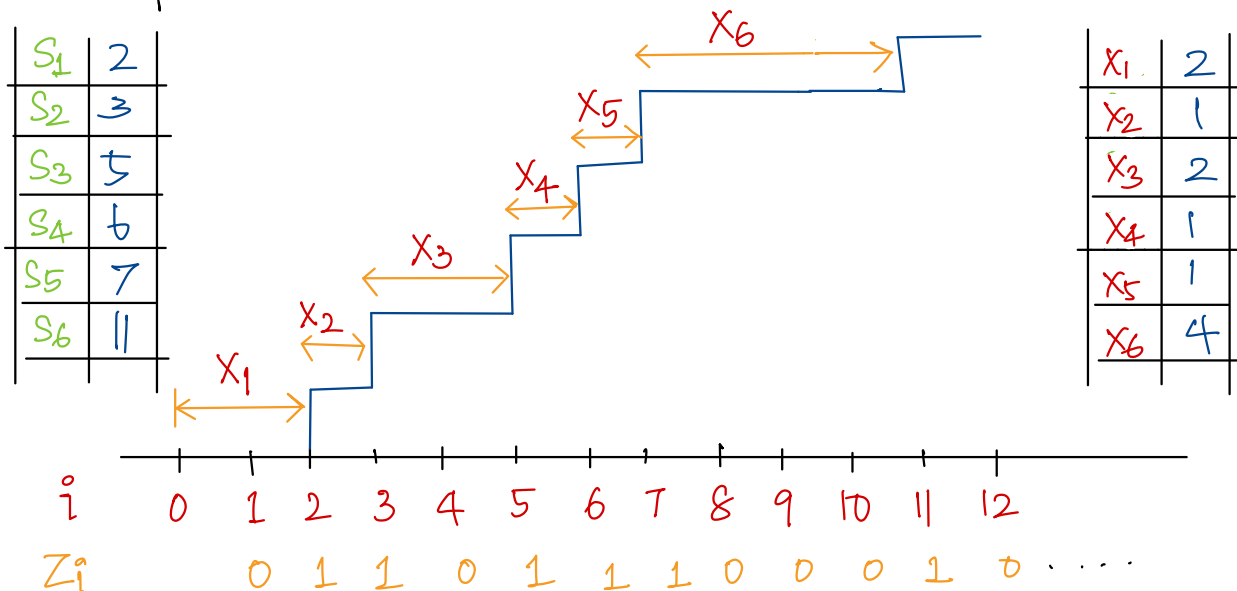
— If the inter-arrival times are IID random variables : Such an arrival process is called a **Renewal process**.

Arrival process: A sequence of increasing RVs $0 < S_1 < S_2 < \dots$, where $S_i < S_{i+1}$
 $\Rightarrow S_{i+1} - S_i$ is a positive RV X s.t. $F_X(0) = 0$. The RVs $S_1, S_2, \dots$ are called arrival instants/epochs and represent successive times at which some random repeating phenomenon occurs (like arrivals, departures, etc).

We saw Bernoulli process $\{Z_i: i \in \mathbb{N}\}$ modeling arrivals at integer values.
 $Z_i = 1 \Rightarrow$ there is an arrival in the epoch i , $i \in \mathbb{N}$.



For this sample path of the Bernoulli process, what is the sample path of the corresp. arrival process and the inter-arrival times?



To fully specify the arrival process, we need to specify the FDDs of the process $\{S_n : n \in \mathbb{N}\}$

Alternate characterizations of the Arrival Process:

i) Sequence of inter-arrival times : $\{X_i : i \in \mathbb{N}\}$: Continuous valued discrete parameter process.

$$X_1 = S_1$$

X_i s are positive RVs

$$X_i = S_i - S_{i-1}, i \geq 1$$

$$S_n = \sum_{i=1}^n X_i \quad \left\{ \text{Arrival epochs in terms of inter-arrival times} \right\}.$$

Thus, the FDDs of the process $\{X_n : n \in \mathbb{N}\}$ also completely specify the arrival process.
In most cases of interest, the inter-arrival times are IID and hence the arrival process is specified using the FDDs of the inter-arrival times.

ii) The Counting Process : $\{N_t : t \in \mathbb{R}^+\}$: (Discrete-valued Continuous parameter process)

N_t - Aggregate number of arrivals in the interval $[0, t]$

Assumption : Arrivals at strictly positive values of t i.e. $S_1 > 0$

$N_0 \triangleq 0$ with prob. 1.

$$N_t = n \text{ for } S_n \leq t \leq S_{n+1}.$$

$$N(t_2) \geq N(t_1) \text{ for } 0 \leq t_1 \leq t_2$$

The arrival process and the counting process are related as :

$$\{S_n \leq t\} = \{N_t \geq n\}.$$

$$\text{(OR) equivalently } \{S_n > t\} = \{N_t < n\}$$

Example: Processes in a single station queue.

Consider the CCD outlet in B-Block. There is only one server.

Customers arrive at random times and "suppose" they wait in line in FCFS order.

$\{A_n: n \in \mathbb{N}\}$: Arrival epochs. $\{D_n: n \in \mathbb{N}\}$: Departure epochs.

$\{S_t^A: t \in \mathbb{R}^+\}$ = Aggregate arrivals in $(0, t]$

$\{S_t^D: t \in \mathbb{R}^+\}$ = Aggregate departures in $(0, t]$.

$\{X_t: t \in \mathbb{R}^+\}$: Queue length process = # customers in the system at time t .

$\{W_n: n \in \mathbb{N}\}$: Sojourn times of the customers. (Amount of time n^{th} customer spends in the system from the instant she arrives until the instant she leaves.)

Renewal Process: A renewal process is an arrival process for which the sequence of inter-arrival times is a sequence of positive IID random variables.

Defn: A **Poisson process** is a renewal process in which the inter-arrival times have an **exponential CDF**, i.e. for some real $\lambda > 0$, each X_i in $\{X_i: i \in \mathbb{N}\}$ has the pdf $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

λ : Rate of the process.

Poisson processes constitute the simplest and the most widely-used class of renewal processes. Why? Because of the **MEMORYLESS** property of the exponential distribution.